Department: Data Science and Artificial Intelligence
Course Name: Data Science Tools and Software
Course Code: 03203N
Professor Name: Dr. Laila M. Fatehy
Level: Two

Sheet 3
(Python Review)

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[1] Solve the following questions using Python :

(1) Given the following NumPy arrays x= [10 20 30 40 50] and
y=[[10 20 30 40 50]
[1 2 3 4 5]] What will be the dimension, size, and shape of x & y ?

(2) If you want to change the shape of the array y from its original shape to (5,2), what will be the new shape of the resulting array

(3) Create two NumPy arrays with the following values:

	[1, 2, 3],		[2, 3, 4],
ar1:	[5, 10, 15],	ar2:	[6, 7, 8],
	[10, 15, 20]		[9, 10, 11]

(4) If you perform element-wise multiplication, division, and addition on the two NumPy arrays ar1 and ar2, what will be the resulting arrays?

(5) If you vertically and horizontally combine the two NumPy arrays ar1 and ar2, what will be the resulting arrays?

(6) Create two NumPy arrays with the following specifications:

- **The first array (ar3) should be a 3x2 matrix filled with zeros.**
- **The second array (ar4) should be a 4x4 matrix filled with ones.**

(7) Create a square NumPy matrix where the main diagonal contains the values 10, 20, and 30, while all other elements are zeros.

(8) Create a NumPy array containing a sequence of integers starting from 0 up to (but not including) 10. What will be the resulting array?

(9) Create a NumPy array containing a sequence of integers starting from 5 up to 9. What will be the resulting array?

(10) Create a NumPy array starting from 2, ending before 12, with a step of 3. What will be the resulting array?

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[2] Solve the following questions using Python :

(1) **Create a Pandas Series that stores the data for the person Saad Mohamed, whose age is 26. What will be the resulting Series?**

(2) **For the given Pandas Series, determine its shape, number of dimensions, size, index labels, and values. What will be the output of these attributes?**

(3) **How can you check if a specific index label ("age", "address") exists in the Series?**

(4) **How can you retrieve the value associated with the index label "fname"?**

(5) **How can you access the values "saad" and 26 from the given Pandas Series using positional indexing & index labels?**

(6) **How can you update the value of "age" in a Pandas Series from 26 to 27?**

[3] Write Python script to calculate the following:

(1) How can you create a Pandas DataFrame from two Series, where one column has more elements than the other?

(2) How can you create a Pandas DataFrame with two columns labeled 'x' and 'y', where the rows have custom index labels 'a', 'b', and 'c'?

(3) Add a new column 'z' with values 100, 200, and 300 to an existing Pandas DataFrame? What will be the updated DataFrame after this operation?

(4) How can you create a DataFrame with a single row containing values x = 4, y = 40, z = 400 and assign it an index label 'd'?

(5) How can you add a new row from another DataFrame (newdf) to an existing Pandas DataFrame (df)? What will the updated DataFrame look like after this operation?

[4] Read data from a CSV file named 'section_data.csv' to solve the following

(1) How can you display the first and last few rows of DataFrame and get a summary statistics for numerical columns?

(2) How can you access single column "height" from DataFrame?

(3) How can you access multiple columns ("height" and "age") from a DataFrame?

(4) How can you filter a DataFrame to show only the rows that contain the maximum and minimum values in each column?

(5) How can you filter a Pandas DataFrame to display only the rows where the age column is greater than or equal to 30?

(6) How can you filter a Pandas DataFrame to display only the rows where age is greater than or equal to 30 and height is less than or equal to 170?

Good Luck
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